

Business Problem Statement

Blinkit Grocery Sales Analysis

Blinkit, India's leading instant grocery delivery platform, aims to gain deeper insights into its product sales, outlet performance, and customer purchasing patterns to improve operational efficiency and maximize revenue. The management team wants to understand how different outlet characteristics, product categories, fat content, and location types contribute to overall sales performance.

They are particularly interested in identifying high-performing outlet types, best-selling product categories, and the impact of outlet size and establishment year on business growth. The company also seeks to monitor key business metrics such as total sales, average sales, customer ratings, and inventory distribution to support strategic decision-making.

You are tasked with analyzing Blinkit's grocery sales dataset to answer the following business question:

"How can Blinkit leverage sales and outlet performance data to optimize inventory planning, improve outlet performance, and maximize overall business revenue?"

Deliverables

1. Data Preparation & Exploration (Python)

- Load and inspect the raw grocery dataset.
- Perform data cleaning and preprocessing.
- Handle missing values.
- Standardize categorical values.
- Conduct Exploratory Data Analysis (EDA).
- Generate statistical summaries and visualizations.

2. Database Analysis (PostgreSQL)

- Import the cleaned dataset into PostgreSQL.
- Design SQL queries to calculate KPIs.
- Perform sales analysis across products and outlets.
- Generate business insights using aggregate and analytical SQL queries.

3. Interactive Dashboard (Power BI)

Develop an interactive dashboard displaying:

- Total Sales
- Average Sales
- Number of Items
- Average Rating
- Sales by Outlet Type
- Sales by Outlet Location
- Sales by Item Type

- Sales by Fat Content
- Outlet Size Distribution
- Outlet Establishment Trend

4. Business Report

Prepare a report summarizing methodology, key findings, and business recommendations.

5. GitHub Repository

Include:

- Python Notebook
- PostgreSQL Queries
- Power BI Dashboard
- Dataset
- Documentation (README)